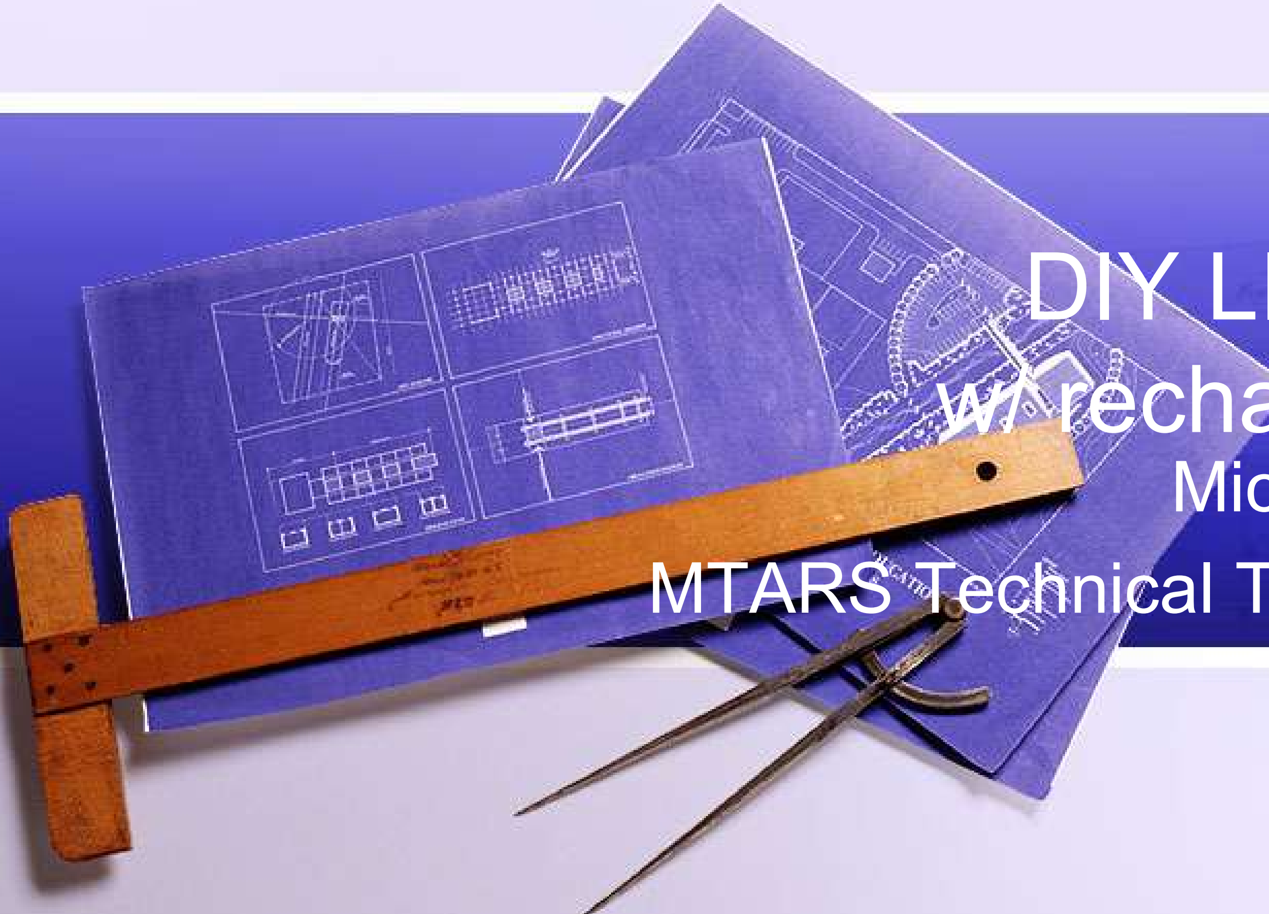

DIY LIPO4FE Battery w/ rechargeable lantern

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MTARS Technical Topic Spring/Summer 2026

Contents

- Disclaimer - Discussion Limited to DIY 15 Amp/hour battery
- Goal establish a basic understanding of building a battery
- References – all images and articles are referenced in the last section
- Bonus material – rechargeable lantern build

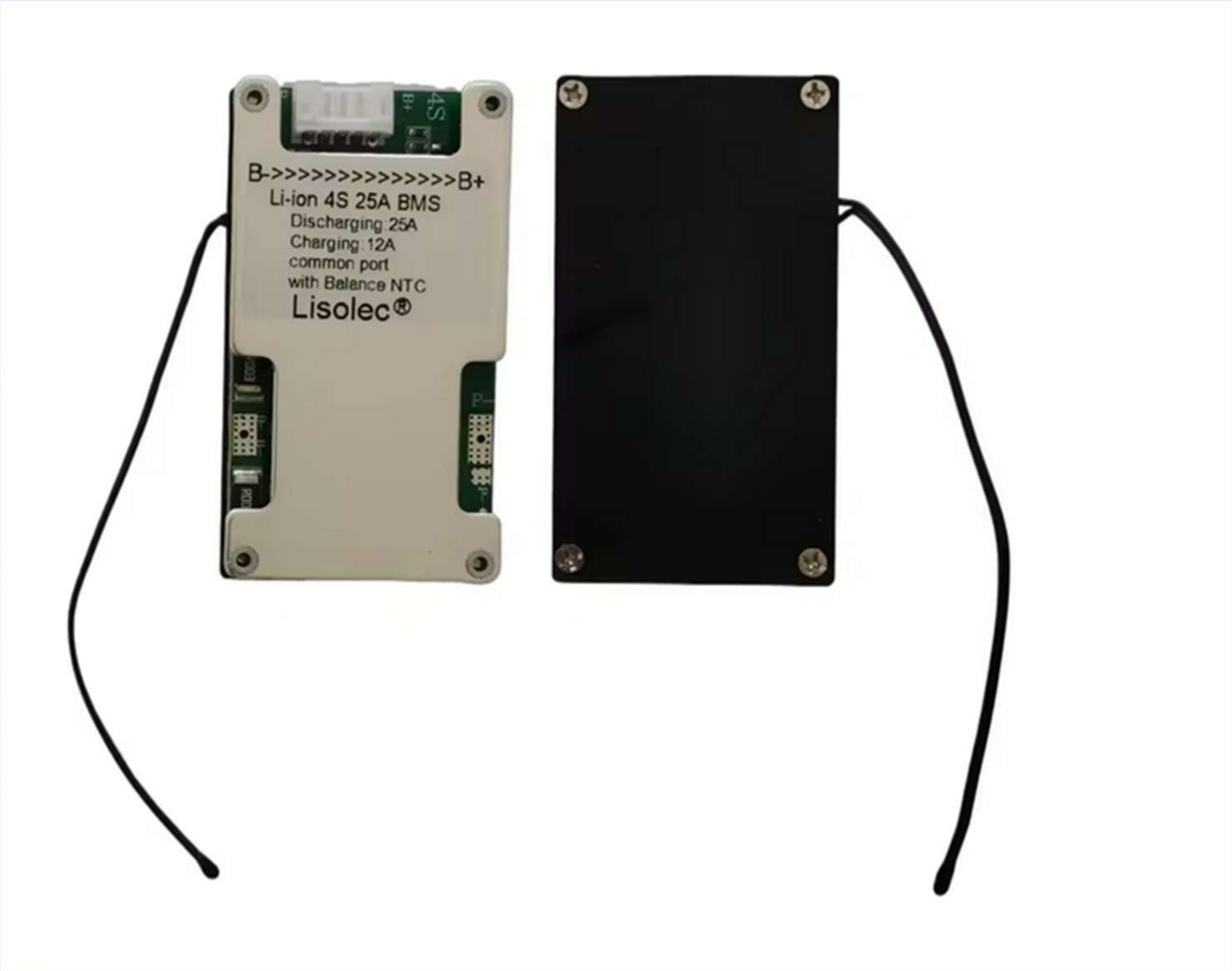
Why DIY: why why why…

- Possible to save Money
  - **Battery as a system – maintainable – replaceable components**
  - **Modular**
  - **Savings in bulk**
  - **Geometry choices –**
    - Standard batteries usually follow prescribed sizes (blocky chunky) Group 31
    - Some flat (truck bed batteries – are better configured for less space $$) duplicate small size 50 amp/hour in cylindrical cells. $140 cells 60mm x 157 mm ~ 2 ½ x 6 ¼”)
      - Amo box case scale
    - Cells available up in to in excess of 100 amp/hour (330!)

DIY Battery - overview

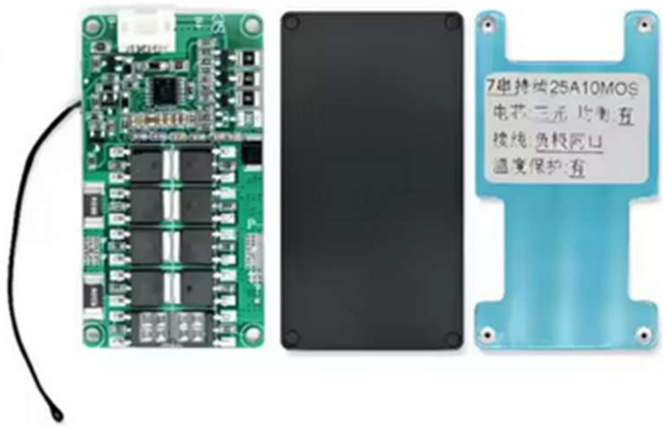
- Lithium Battery Basics
  - **Battery Cells (nominally 3.2 volts)**
  - **Requires a BMS circuit (Battery Management System) – single point of failure**
    - Lithium cells require care and feeding , over temperature protection, overload, under and overvoltage
    - Active Component devices – regulated between 14.4 – 11.9 volts nominal V
    - Voltages (Nominal) 12, 24, 48, 72 volts
      - Base 4 (4 battery /12 volt stepsl)
    - High voltage systems possible
  - **Extended Life 4000 cycles**
  - **Chemistry (Lipo4FE) Lithium Iron –**
    - Less prone to thermal runaway
    - Vent hydrogen when severely abused – but don't explode or catch fire on their own..

BMS: Lisolec - $0.99 new shopper deal – smaller than a deck of cards – 20 amp rating



BMS Specs

20A BMS
wiring:Common Port
Continuous working current:20A
Charging current:10A
Product Size:60*33*7MM
connecting cable :Yes
charging protection: Yes
Discharge protection: Yes
Short circuit protection :Yes
Wire break protection :Yes
Battery Balanced :Yes
Temperature control protection:Yes
Please read the webpage content for detailed parameters.



(4) C40 Cells 20 Ah ~$36

Diameter: 1 – 9/16”
Length: 4 -15/16”

Accidently purchased 15 amp/hours ~ same cost

Construction

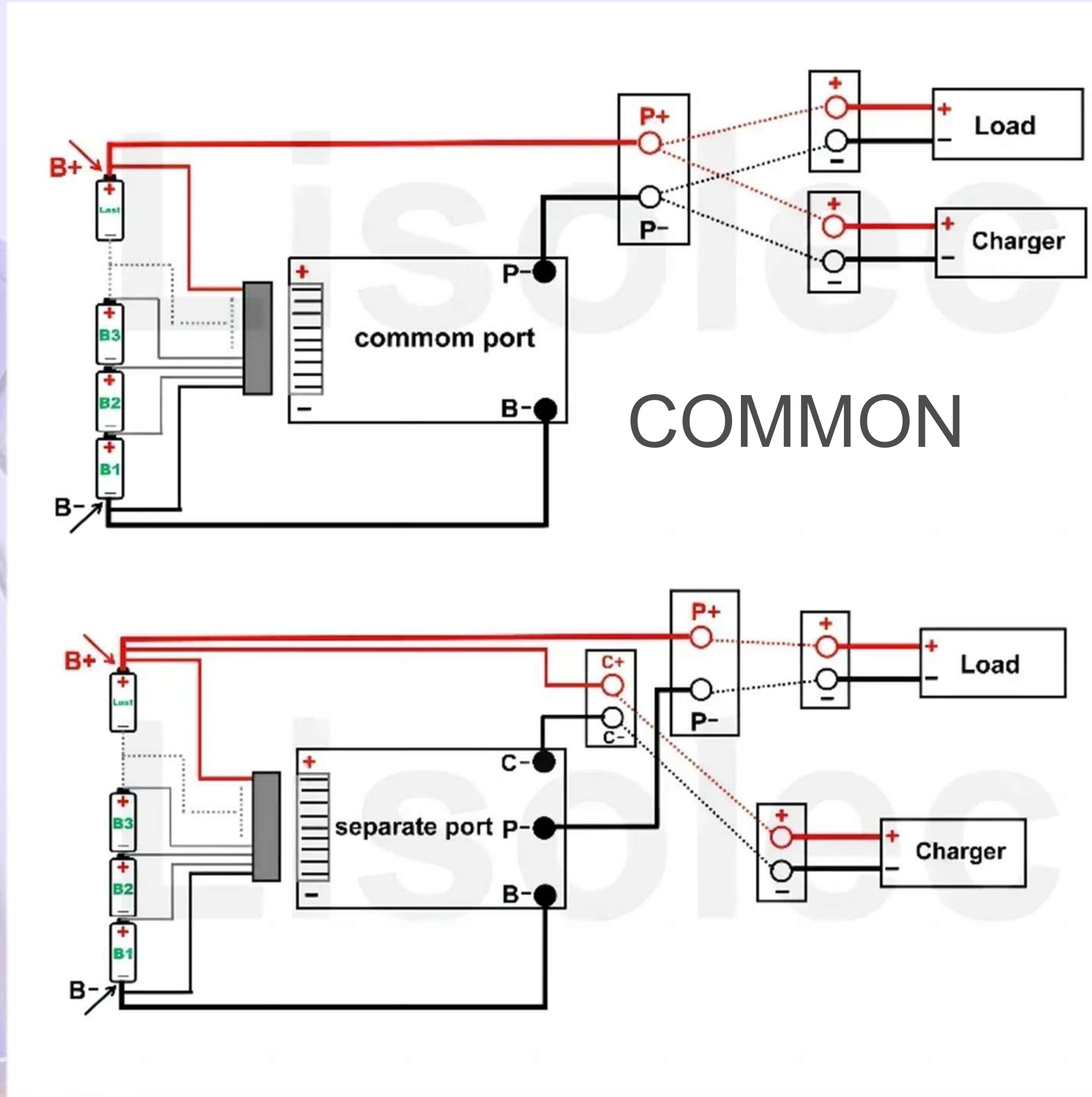
- Battery to battery Bus Bar or jumpers
- BMS connections ring terminals (user installed)
- Batter to load and charge – personal choice
  - **Anderson 30 Amp**
  - **Barrell connector (charge)**
- Housing Amo box or Harbor Freight case (depending on the size)
- Easily integrated into Power Box
- Tools needed – ¼” torque driver (connections should be torqued to spec)
- Secure with insulating tape (Kapton type)

Typical 30 amp/hr cells - $62.54

50 amp/hour – tandem terminals

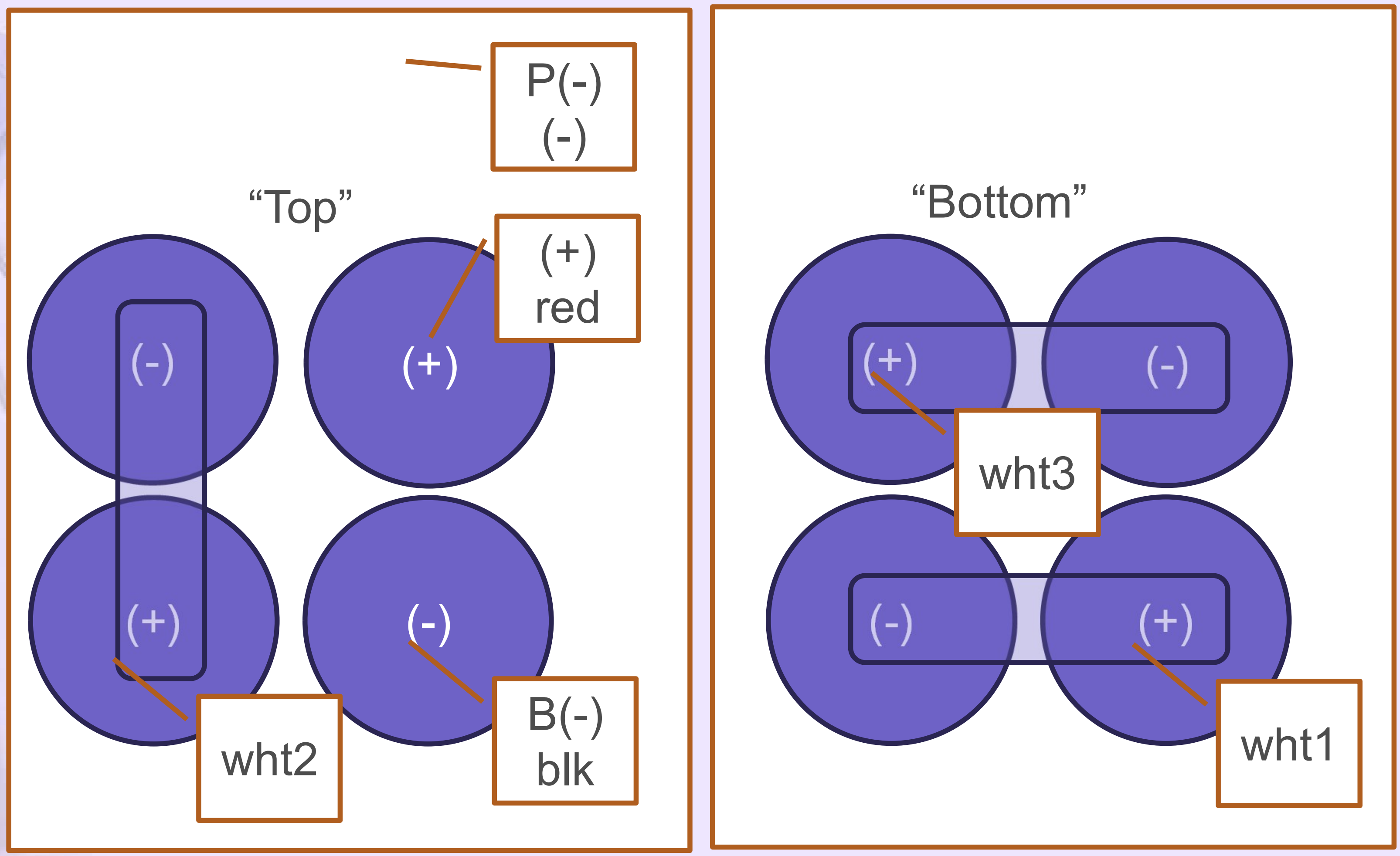
Easy to short out terminations – build with care!
$107 on sale

BMS wiring (12 volt nominal)

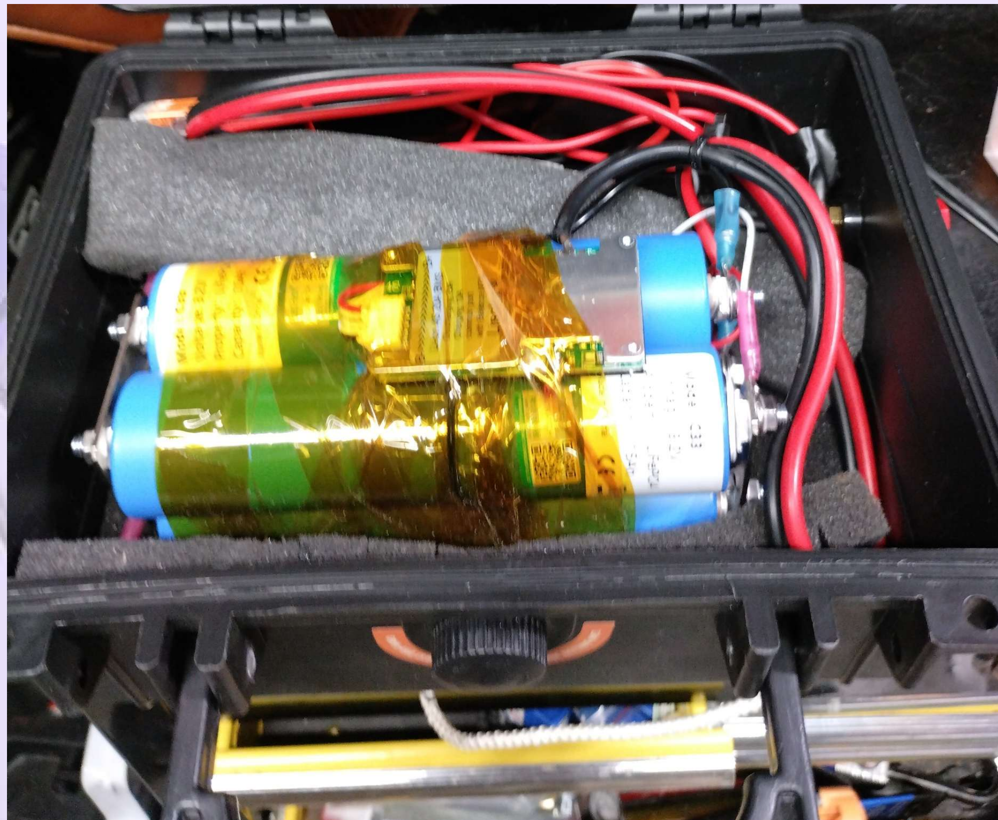


Battery Arrangement

Battery connection to (+) battery plus (also charge + ; (-) power to P(-) along with charge (-)



Installation into Small Case

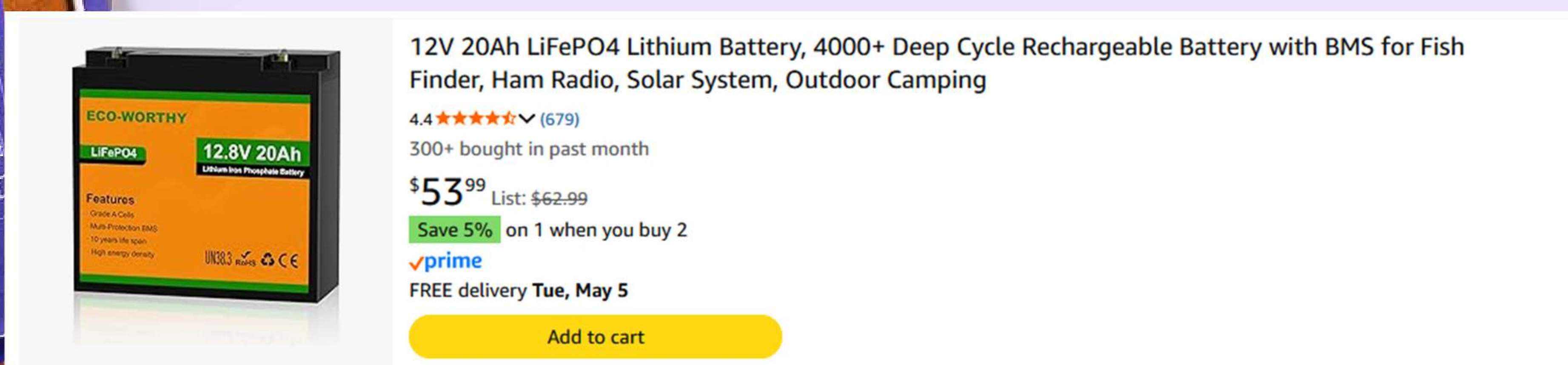


Comments

- Special thanks to Will Prouse – YouTube Channel and other contributors
- Solder required for B-, and P- to BMS lots of heat and tin wires
- Assembly is cumbersome
- Build with care – bus bars can flop around
- Built flat then rotated over – mindful of shorts – no off switch
- Kepton tape used to secure batteries, leads and BMS
- Wire tie secure the larger wires and reduce stress on solder joints. (B-, P-)
- Working – charging from Lipo4fe charger.

DIY cheaper???

- Cost Savings at 20 Ah??? Cells $34 +bms $11 ..~$44
- 20 Ah Eco-worthy is $53!..
- Higher Ah and multiple cells required to save very little . 4 prismatic ~100 amp/hour (- $149.50)
- No Case
- Advantage maintainable?



Questions / Comments?

- On to bonus topic.

Bonus Topic DIY Lighting addendum.

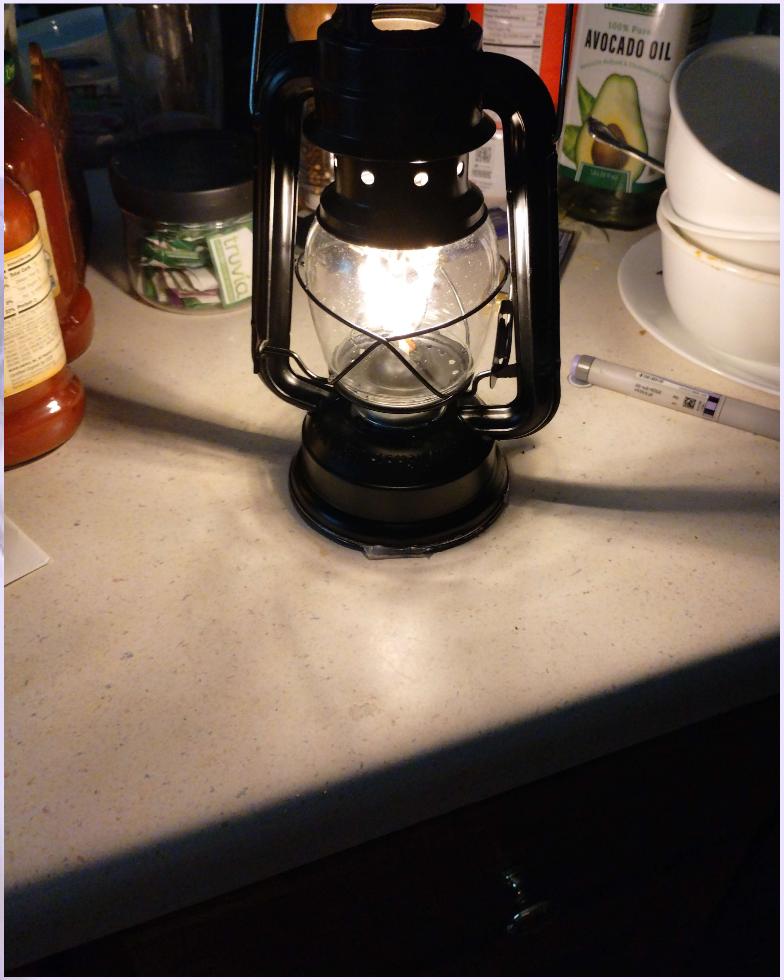
- When 16 slides aren’t enough.
- Earlier Jar lights (12 volts) require a power source
- Rudimentary
- First attempt self contained – glass jar with speed control, single filament and dry cell pack .. Meh.. (great prototype)

Rechargeable Lantern

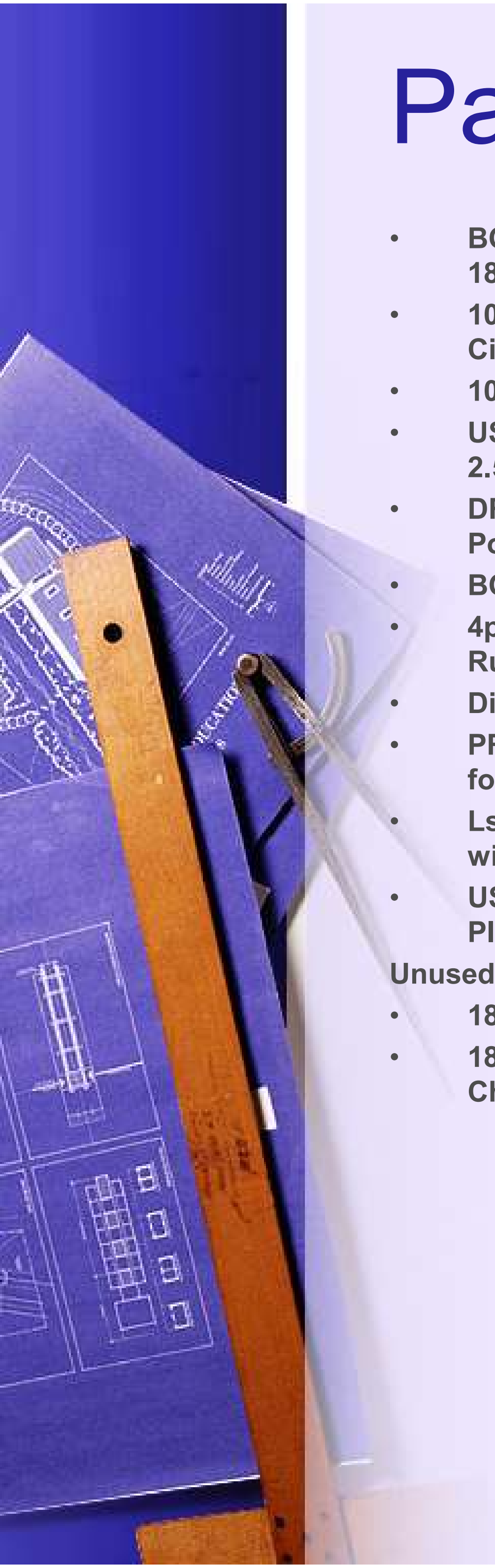
- Inspired by Ufbaehs Projects
  - https://www.youtube.com/@ufbaehs08
- Based on Standard filament 3.0 v led

Lamp in use – thunderstorm – power outage.

Lantern Upgrade: (better prototype) Self-contained



- Repurposed Oil lamp
- 6 to 32 volt input
- 3.7 lipo battery (3000 ma/hours)
- Adjustable brightness (impacts run hours.
- Rugged – no flame.
- Runtime ~ 3hours
- Run while charge (days) – direct solar rechargeable.



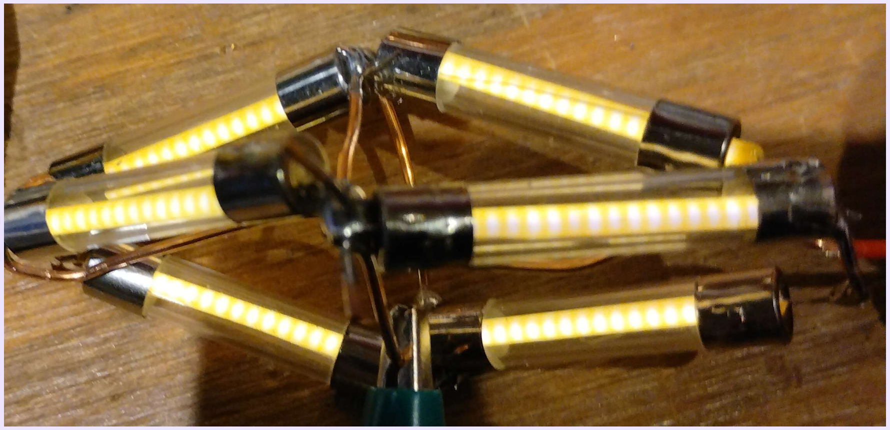
Parts – extensive BOM

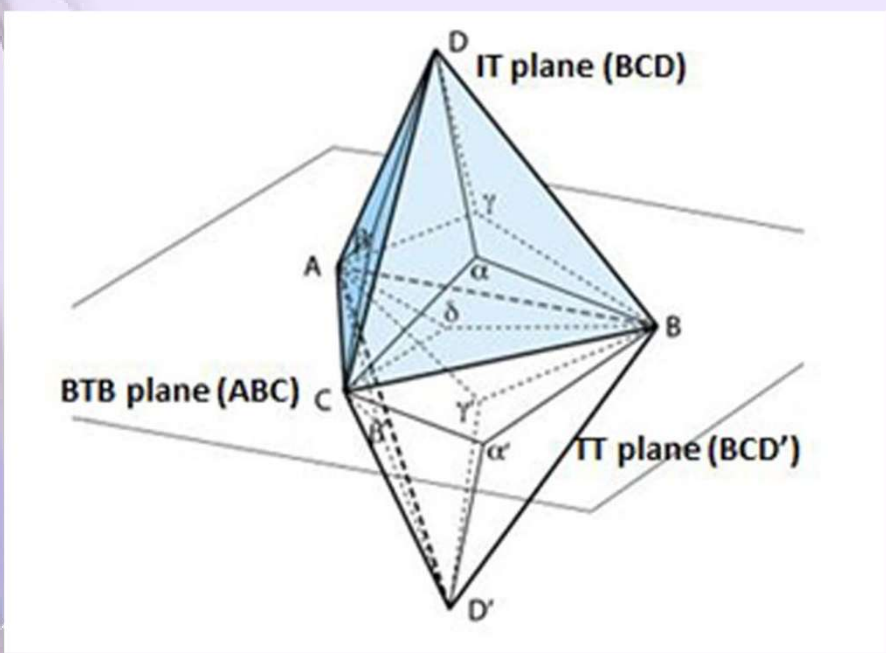
- BOJACK Low Voltage DC Motor Speed Controller PWM 1.8v 3v 5v 6v 12v 2A 30W Adjustable Driver Switch 1803BK 1803B
- 10 Pack Ultra-mini USB Type C 3.7V Lithium Battery Charger Board 4.2V Charging Module with Protection Circuit and LED Charge Indicators 5V USB-C Input
- 10 Pairs 1.25 mm JST 2 Pin Micro Electrical Male and Female Connector Plug with 80mm Wire Cables
- USB 2.0 to Bare Wire Cable,USB Type A Male Plug to 2 Pin Open End Bare Wire Pigtail Power Cord,24AWG 5V 2.5A 2 Cores Power Extension Cable for LED Strips, DIY Projects Electronics Repair-0.3M (5 PCS)
- DROK USB Buck Converter, 4pcs DC-DC Step Down Module 6-32V 12V 24V to 5V QC 3.0 Charging Module Power Supply Voltage Regulator Volt Transformer Board
- BOJACK 6x30mm 0.5 A 0.5 amp 250 V 0.24x1.18 Inch Volt F0.5AL250V Fast-Blow Glass Fuses(Pack of 18)
- 4pcs 3.7V 3000mAh Lithium Polymer Rechargeable Battery 1S LiPo Battery with Protection Board Insulated Rubber Tape and JST 1.25 Plug for ESP32 Development Board, Mesh Node T114
- Diode
- PRETYZOOM 100pcs LED Filament Replacement Parts, 3V DC 120MA 38MM, Warm White 2700K Hard Diodes for Solar Garden Light Repair, DIY Retro Edison Bulb Projects & Low Voltage Fixtures
- Lsgoodcare 20Pcs 5.5 x 2.1 MM DC Power Jack Socket Dc099 30V Threaded Female Mount Connector Adapter with Cable & 5.5mm x 2.1mm 12V Pure Copper Male DC Power Pigtail Cable
- USB Type C Female Socket to 2 Pin Terminal Cable 3.5inch, USB C Female Power Connector with Dustproof Plug, 5V/2A USB Panel Mount USB Type C Power Port

Unused: (SIZE CONTSTRAINTS)

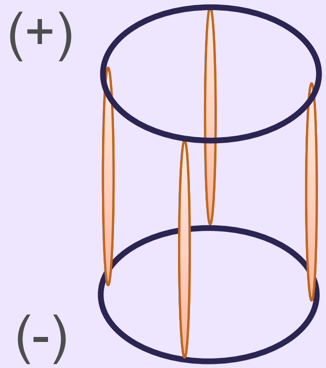
- 18650 Battery Holder Li-ion Battery with Dual Compartment Support Micro USB 5V/3A 3V/1A V8 Battery Shield
- 18650 Battery Holder Li-ion Battery with Single Compartment Support USB 5V/2A 3V/1A V3 Battery Shield Charging Module with USB to MicroUSB Charging Cable

Filament assembly – two-sided Tetrahedron

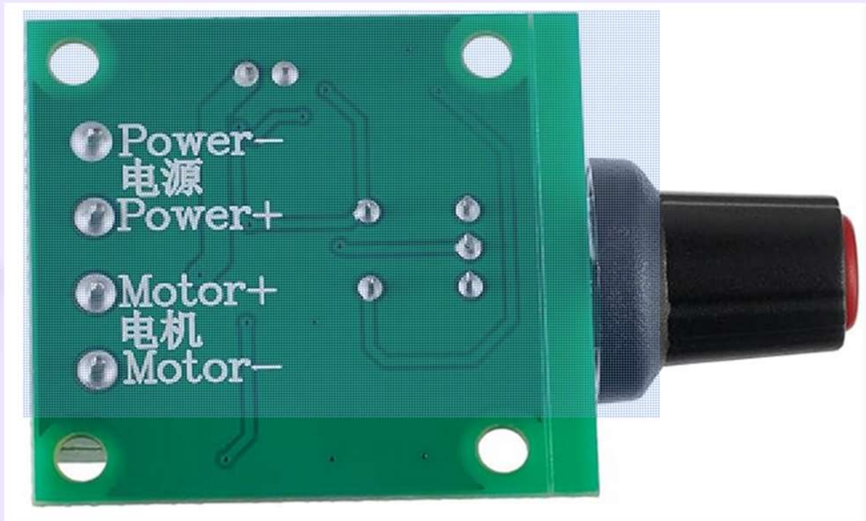


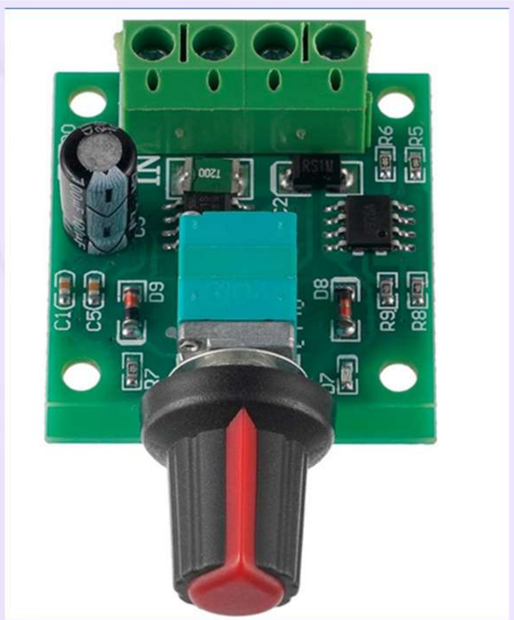


- Current draw ~600 mA at 3 volts.
- Solid Copper wire used for build
- Difficult to solder
- Mounting/wiring in lantern – wire tie through wick
- Filaments mounted in glass fuses
- Tried simpler geometry – mounting issues (fragile filament only)!.



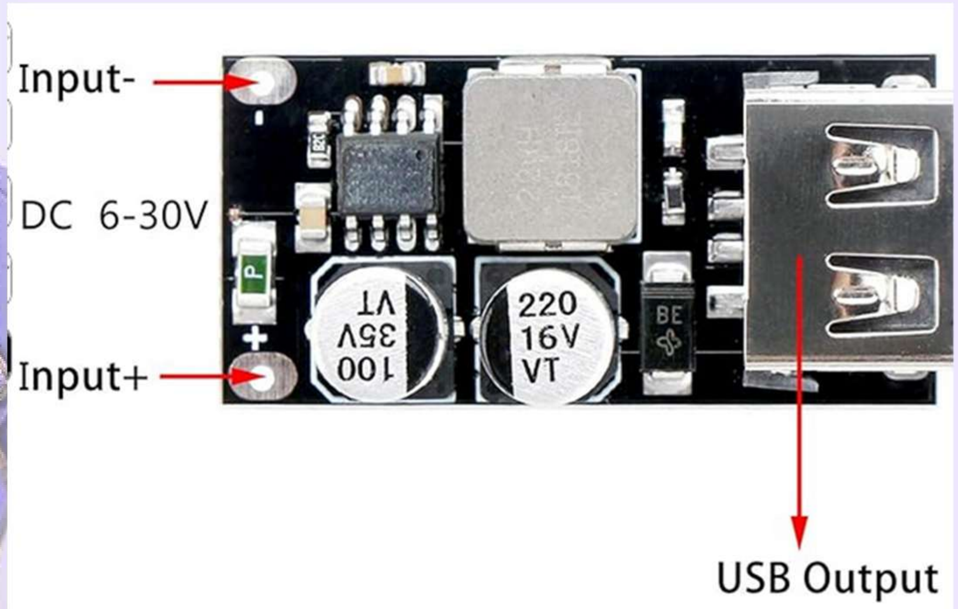
Speed controller





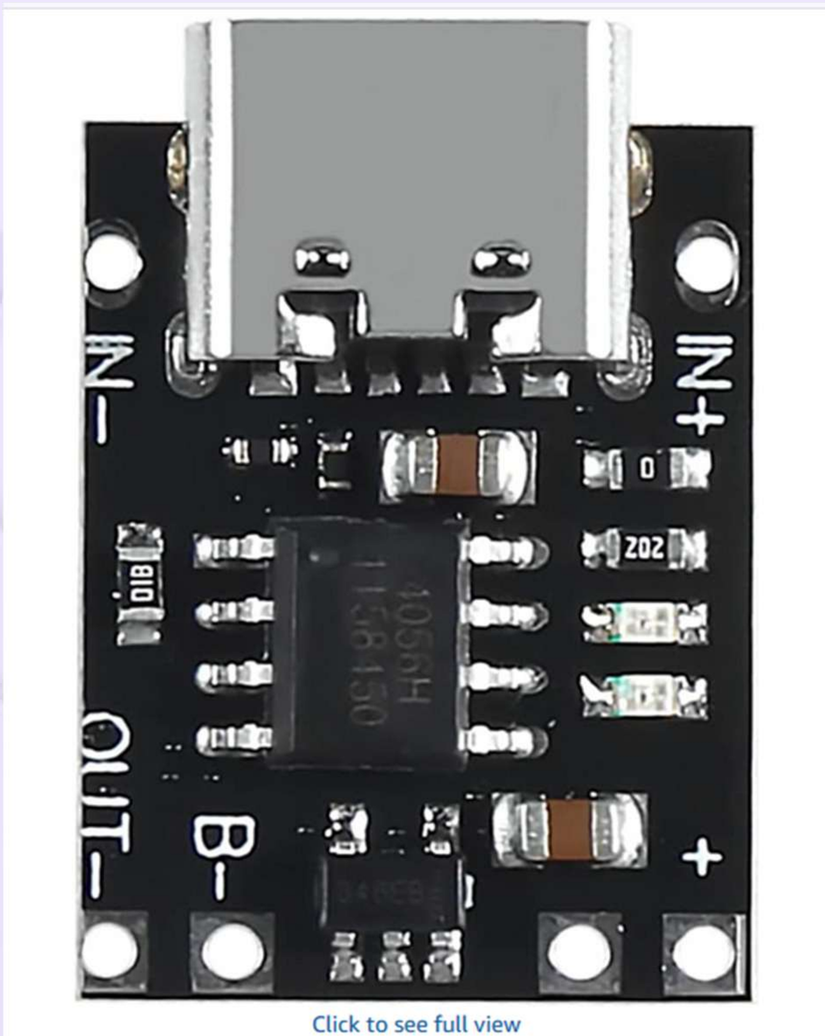
- 3.7 volts in
- ~ 3.7 volts out
- Diode used as step down
- PWM based
- Switch On/Off
- Works great as a dimmer.

DC to DC controller -



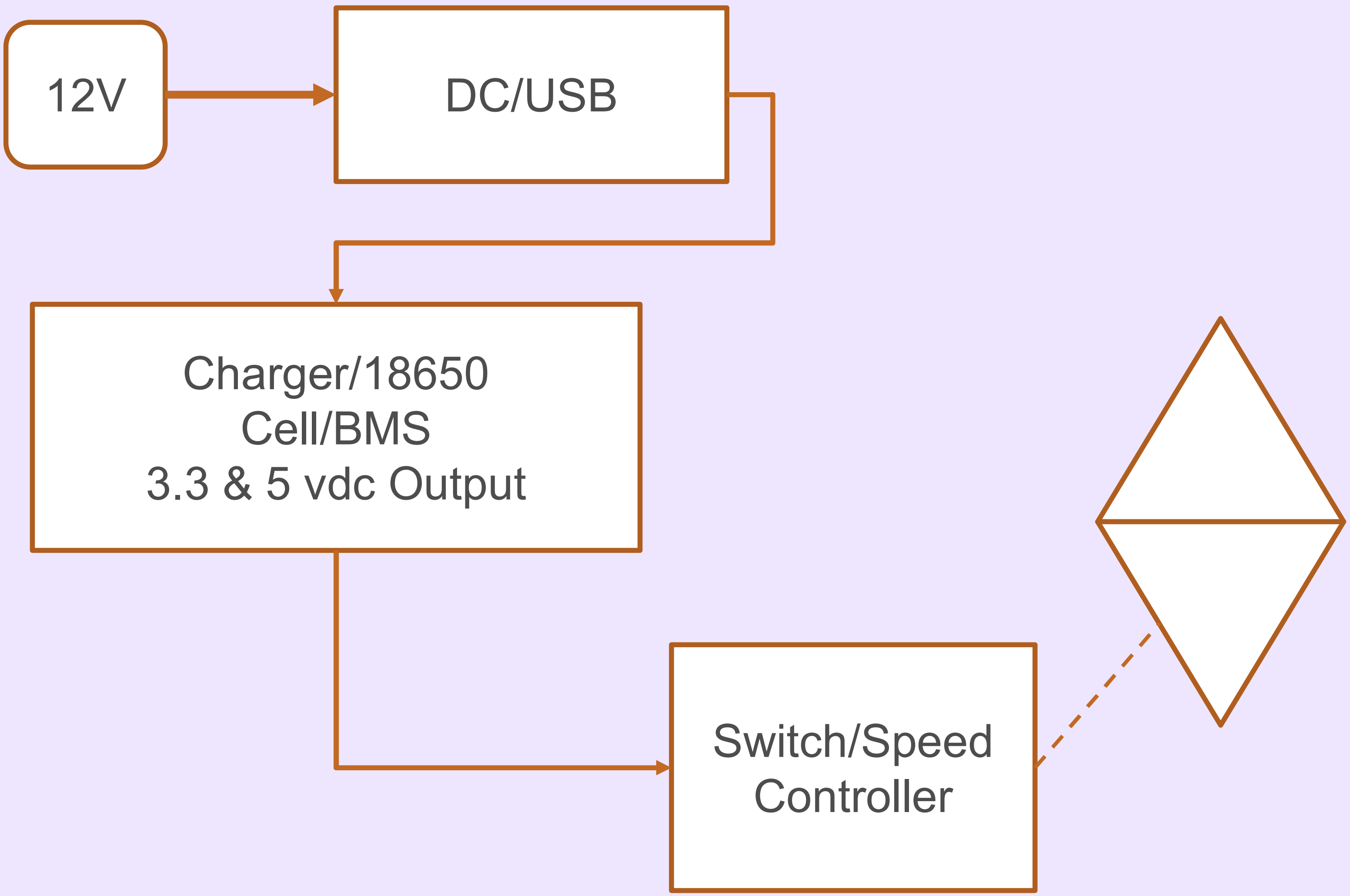
- 12 volts input,
- USB 5 volts out.
- Diode protected against reverse polarity.
- +/- pigtail wago connection to barrel leads.

Battery Charge controller (bms)

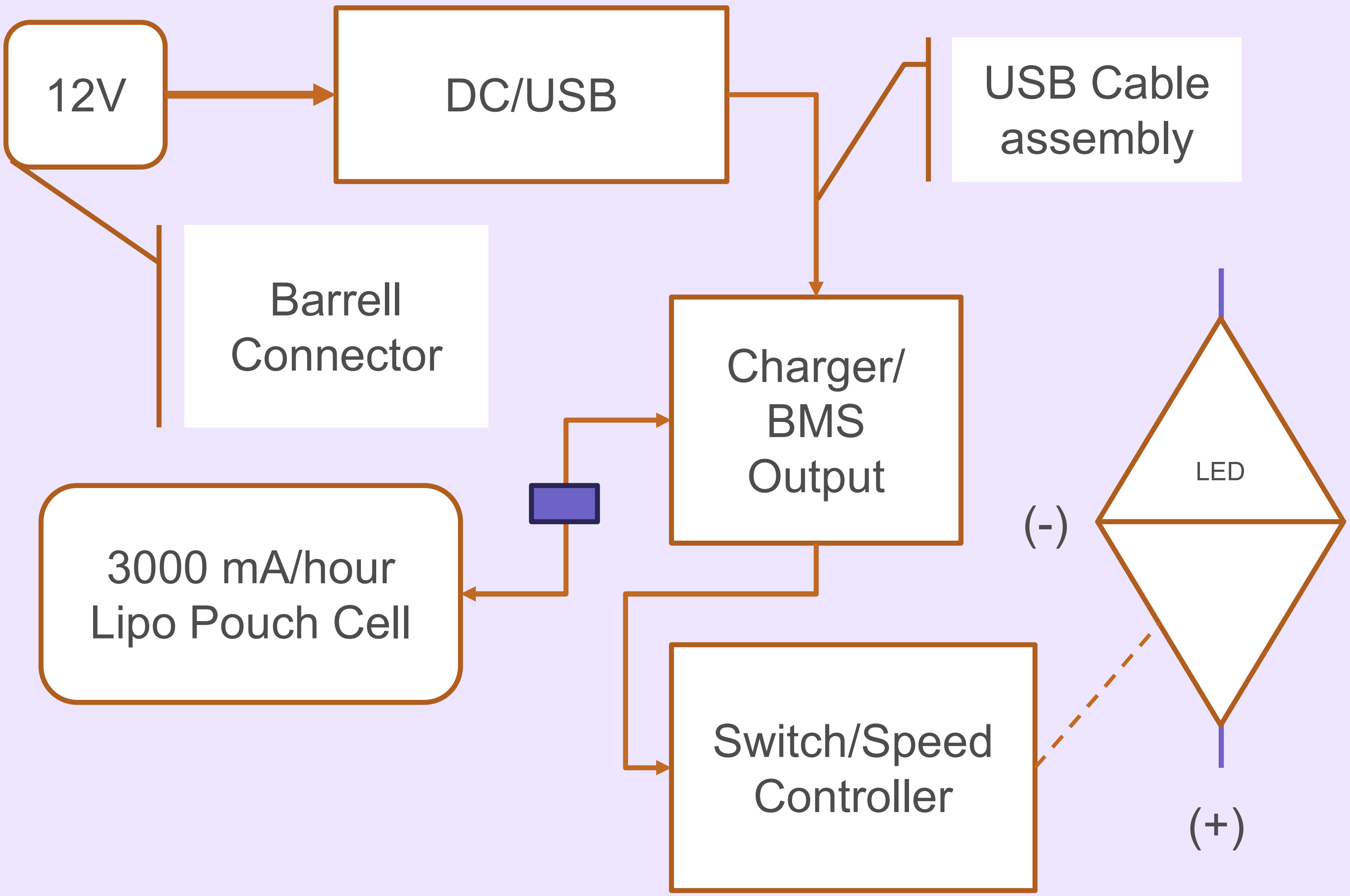


- USB –C input or direct wire to 5 volts
- Rated 1 amp current,
- Charging causes it to warm.
- Difficult to insulate – heat shrink.

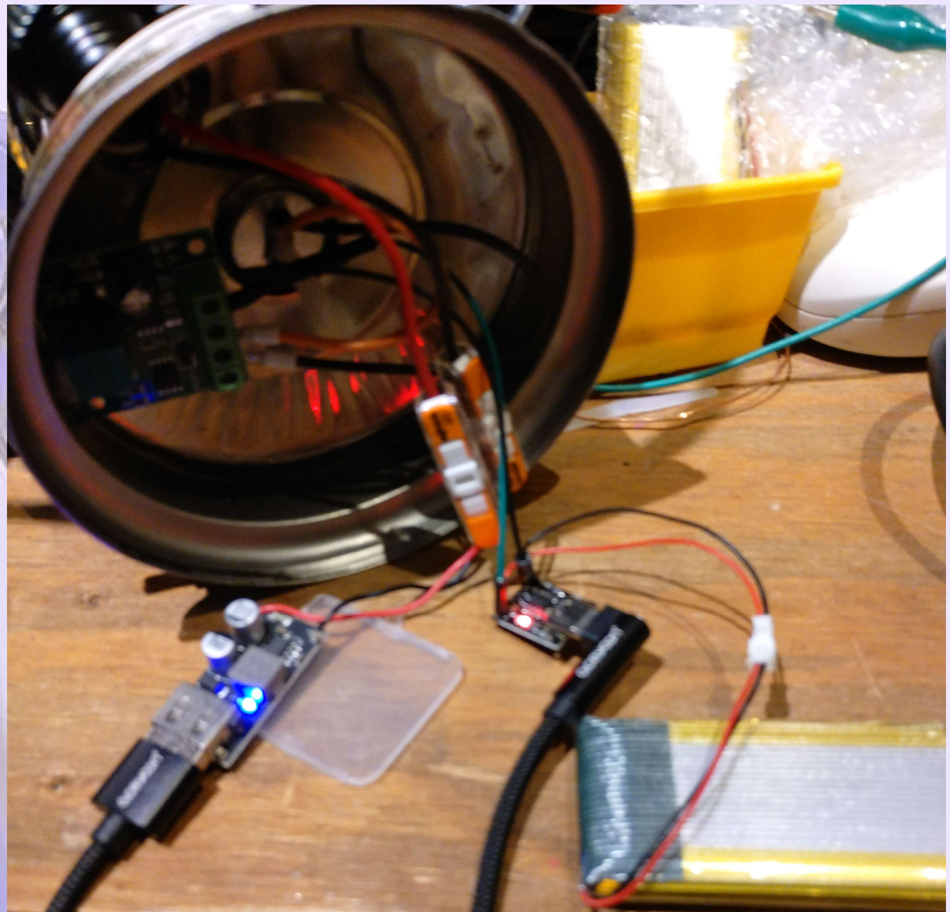
Wiring: (Block diagram style) – attempted – abandoned due to space limitations – suitable for larger project housings. (holes in the bottom of the lantern are artifacts for mounting the 18650 holder.



Wiring: (Block diagram style) – used more parts but compact



Final assembly -



- Packet style cell
- Charge controller
- 12 volt to USB power
- Cell bms/charger
- Speed controller
- USB to USB-C jumper used (short)

Assembly notes 1 of x

- ID can-opener – Pampered chief can opener cuts the bottom of the lantern out leaving an easily reusable bottom.
- Drill hole for speed control/switch
- Drill hole for barrel connector (oil cap/secure with hot melt glue)
- Element is feed through the wick opening with short +/- leads
- +/- element leads motor terminal +/-
- Barrel connector leads wago to DC converter input +/- [solder pigtail – small wire] (check polarity!)

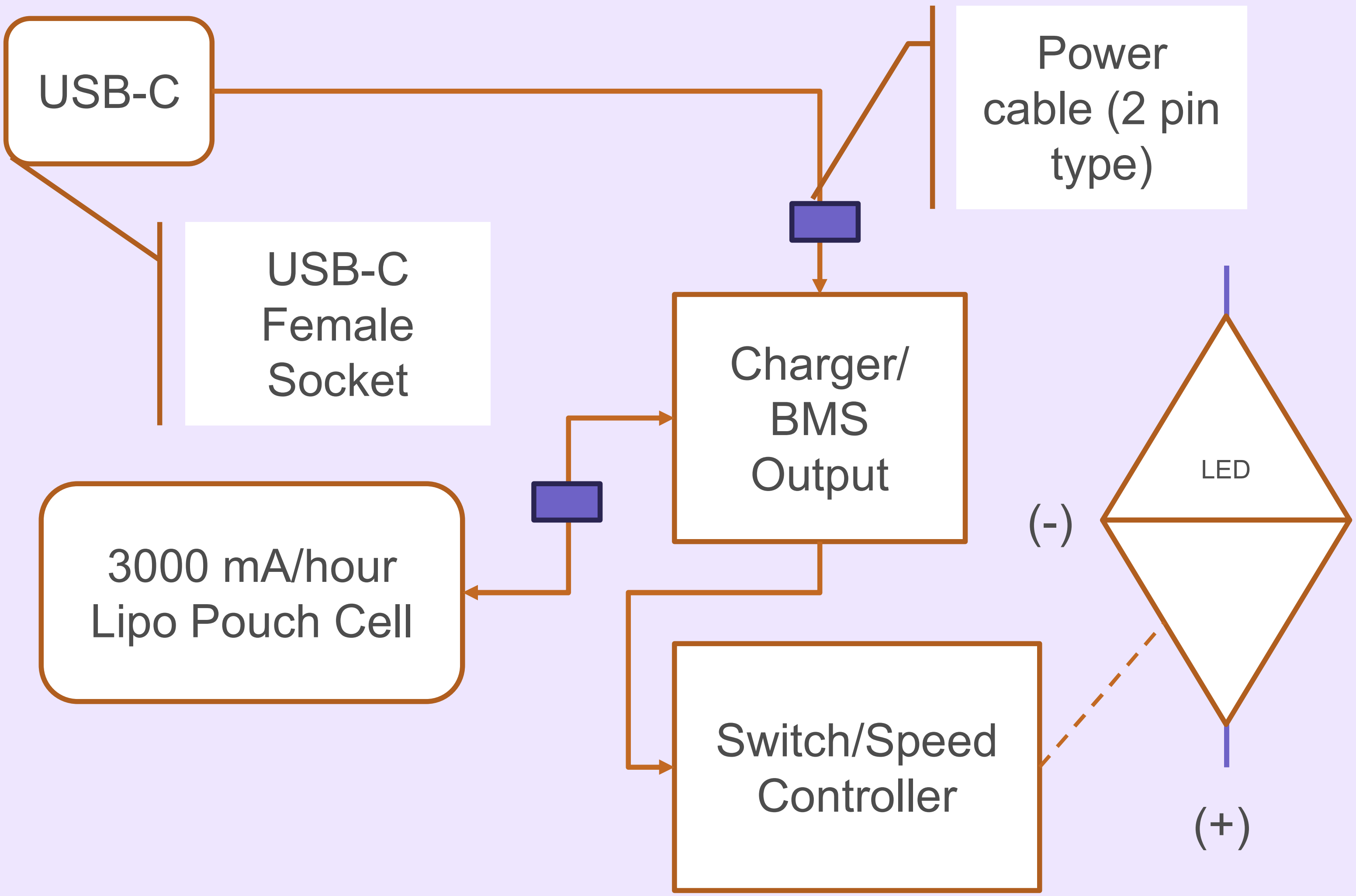
Assembly notes 2 of x

- Connect USB jumper or use cable assembly.
- 5 VDC to USB-C or +/- Input leads of battery charger/bms (base of DC/DC glued to plastic piece)
- Charger BMS Solder small wire (PC twisted pair leads) to charger +/- Out wire to +/- Supply of speed controller – I used Diode to drop it 0.7 volts – (necessary?)
- Solder small leads to BMS output pigtail.
- Charger BMS solder female battery connector.
- LED assembly wago clamps to LED assembly
- Plug battery to cable
- Arrange boards and cables
- **Test, then hot melt glue base back on – work around pressing back together using a heat gun**.
- Battery replacement requires a heat gun to separate bottom (iphone technique)

Lamp version 2

- Eliminate 12 volt to USB adapter
- Utilize USB-C feedthrough
- Mount heatsink to charger/bms (glue upside down to baseplate with epoxy (heat sink)
- Solder wires to charger/bms dead bug style.
- Cover circuit back with liquid electrical tape.

Wiring: USB charged v 2– even fewer parts



USB-C Weatherproof Plug Approach

- Skips 12V/USB converter
- Charges from USB cable
- Sits in fuel cap spot. (drill out)
- Direct solder to charger/BMS
- 2 pin plug (battery cable type)

Summary

- Harder to build than expected – respect for consumer products. (building another v2)
- Lighting assembly especially challenging to solder. Could have used a jig – next build form with play dough! (remove when finished)
- Very satisfying and easy to use – just turn on when needed.
- Not sure what the battery loss will be when not in use. (battery is connected to bms and 3.7 v supplied to speed control at all times, speed control has an on/off switch. (phantom loss?)

Lamp Project comments/Discussion

- Lamp is irreversibly converted to electric – holes/ bottom of tank removed then glued on
- Another oil lamp project allowed removal and reuse as oil lamp.

References:

- https://poweredtemplate.com/02512/0/index.html
- www.aliexpress.us
- https://www.youtube.com/@WillProwse
- https://www.youtube.com/@ufbaehs08